

SMART INDIA HACKATHON 2026



- Problem Statement ID- **SIH 2026 (to be filled)**
- Problem Statement Title- **Crowdsourced Road Defect Intelligence for India**
- Theme- **Smart Automation**
- PS Category- **Software**
- Team ID- **To be filled**
- Team Name- **localhost**





The problem we are solving

Indian cities have no automated, real-time way to find and verify road defects. They rely on slow citizen complaints, unverified contractor repairs and costly survey vehicles that reach under 2% of the network.



Our solution

gig-delivery phones (Zomato, Swiggy, Rapido) become a free, real-time road inspection network



What makes it different - and what nobody else does

Zero new hardware: Rs 3, not Rs 17,693 Rivals buy survey vans or bolt-on IoT kits. We ride phones already on the road.	A cost-gated ladder, not one model Free sensors screen every metre, agreement kills false alarms, video confirms the last 5%.	1 wrong flag in 9 becomes 1 in 250 Four phones agreeing on one 20 m segment - maths, not costlier hardware, buys the accuracy.
We train on the look-alikes A quarter of our dataset is speed breakers, manholes and tar patches - where models fail.	A repair proves itself Fixed roads go quiet, and before/after evidence makes contractor payment auditable.	Tuned for Indian two-wheelers Per-phone and per-vehicle calibration; severity weighted by who actually dies.



TECHNICAL APPROACH



How the data moves



What we are building it with

On the phone

React Native (Android and iOS), 100 Hz accelerometer and GPS, on-device 1D-CNN bump classifier (Keras), SQLite queue for offline trips

Our server

Python (FastAPI), Celery workers on Redis, REST/JSON APIs, JWT auth, Pydantic validation

Map and database

PostgreSQL 16 with PostGIS, OSRM for map-matching each reading to a real road, DBSCAN to group nearby hits, GeoJSON output

The AI models

YOLOv11s (Ultralytics, PyTorch) on the clips, OpenCV frame handling, trained on RDD2022

The dashboard

HTML, CSS, JavaScript, React, deck.gl and MapLibre GL for the live map, Chart.js for ward-level trends

Running it

Docker + docker-compose, GitHub Actions, Nginx



FEASIBILITY AND VIABILITY



Why the math works

Wu, Wang, Hu et al. - Sensors 20:5564 (2020)

88.5% of one phone's flags are real. Their published counts also give 1.5% false flags per clean pass.

One phone on its own: 1 flag in 9 is wrong.

We wait for 4 phones to flag the same 20 m.

1 in 9 -> 1 in 250 once 4 phones agree

We credit each extra phone with 3x, not the 50x the paper implies.

So about 1 confirmed spot in 250 is wrong, and the camera check settles those.

Is it actually doable?

YES - every piece already exists and is free

- Every piece is free and open source
- Runs on an ordinary Rs 8,000 Android phone
- One bus route or municipal fleet is enough to start
- Six of us: one app, one server, one dashboard

What could go wrong

Phones kill background apps

GPS drifts by 20-30 m

Speed breakers look like potholes

Video can catch faces

IMPACT AND BENEFITS

What changes on the ground

6+
DEATHS A DAY

pothole crashes, MoRTH

Daily
ROAD CHECK

instead of once a year

100%
REPAIRS CHECKED

evidence before and after

Who	Today	With SETU
Citizens	no warning, nobody answerable	safer roads, repairs on a clock
Municipality	costly audits, guesswork	a ranked repair list, daily
NHAI / PWD	one survey pass a year	every lane that gets driven
Gig workers	vehicle damage, back pain	warnings ahead, no extra work

What it costs to confirm one pothole

Survey vehicle or manual audit

Rs 17,693

Citizen complaint app

cannot be verified

SETU

Rs 3

5,900x

cheaper for the same kind of proof, because the camera only runs when the sensors are unsure.

Side benefits we did not plan for

- Monsoon damage shows up in days
- Less fuel, fewer suspension repairs
- Proof of which contractor delivered



RESEARCH AND REFERENCES



Peer-reviewed foundation

- **Sattar, Li & Chapman - Sensors 18:3845 (2018)**

A phone's motion sensor can spot road damage; sampling rate is the real limit, so we sample at 100 Hz.

mdpi.com/1424-8220/18/11/3845

- **Wu, Wang, Hu et al. - Sensors 20:5564 (2020)**

88.5% precision and 75% recall per 10 m window - the base numbers our four-phone agreement math is built on.

mdpi.com/1424-8220/20/19/5564

- **Sensors 20:409 (2020) - participatory sensing**

Measured GPS error of 27-32 m at speed, which is why we group inside 20 m after map-matching.

mdpi.com/1424-8220/20/2/409

- **Arya, Maeda, Ghosh et al. - RDD2022 (arXiv)**

47,420 labelled road images from 6 countries including India - our training base for the vision model.

arxiv.org/abs/2209.08538

Government data and Indian field proof

- **MoRTH - Road Accidents in India**

9,438 pothole-related deaths in 2020-24, up 53% in five years.

morth.nic.in/road-accident-in-india

- **NHAI Network Survey Vehicles**

The survey fleet covers ~20,933 km of highway about once a year, at high cost. We cover every lane driven.

nhai.gov.in

- **TrafficSense - Mohan, Padmanabhan & Ramjee (MSR, 2008)**

Phones detected potholes and bumps on Bangalore roads 17 years ago - the idea is field-proven in Indian traffic.

[microsoft.com/research - TrafficSense \(MSR-TR-2008-59\)](https://microsoft.com/research - TrafficSense (MSR-TR-2008-59))

- **NITI Aayog Frontier Tech Hub - RoadMetrics**

An Indian phone-camera road-mapping startup: 50,000+ km mapped, adopted by Chennai. The market is real; we make it free.

frontiertech.niti.gov.in - RoadMetrics story